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Device and method for beading a tube end

Abstract

A device (10) for beading a tube end (12) has a receptacle unit (20) for the workpiece (14) to be beaded and at least one beading die (30) which is mounted in a longitudinally displaceable manner and is aligned with the receptacle unit (20). A protective film (40) is positioned between the beading die (30) and the receptacle unit (20) for the workpiece (14) to be beaded. In a method for beading a tube end (12) first the workpiece (14) to be beaded is positioned in a receptacle unit (20). Subsequently, a protective film (40) is positioned between the receptacle unit (20) and a beading die (30) which is mounted in a longitudinally displaceable manner, before the free edge (12) of the workpiece (14) to be beaded is beaded inwards by means of the beading die (30).

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the benefit of European Patent Application EP 24 162 284.4, filed on Mar. 8, 2024, the content of which is incorporated in its entirety.

TECHNICAL FIELD

[0002] The disclosure relates to a device and a method for beading a tube end. Here, the edge of a metal sheet or a tube end is bent over by means of a corresponding device or manually. In this way, for example, sharp cut edges can be prevented. It is also possible to close a tube end by means of beading.

BACKGROUND

[0003] Devices for beading a tube end are generally known. As a rule, devices of this type have a receptacle unit for the workpiece to be beaded. This receptacle unit can be rotated about a first rotational axis. Moreover, there is a tool holder, to which at least one beading roller and at least one bead roller are fastened via a common axle. The at least one beading roller and the at least one bead roller can be rotated about a second rotational axis along the longitudinal axis of the common axle. The longitudinal axis of the common axle and the rotational axis of the receptacle unit are oriented parallel to one another. A device of this type is known, for example, from U.S. Pat. No. 8,495,901 B2.

SUMMARY

[0004] The present application discloses an improved device and an improved method for beading a tube end, with the result that beading with as low wear as possible can be achieved.

[0005] The device for beading a tube end has a receptacle unit for the workpiece to be beaded, and at least one beading die which is mounted in a longitudinally displaceable manner and is aligned with the receptacle unit. A protective film is positioned between the beading die and the receptacle unit for the workpiece to be beaded. As a result of the film, merely minimum friction occurs between the beading die and the tube end to be beaded, with the result that virtually no abrasion is to be observed. Therefore, the high surface quality of the tube end to be beaded can be maintained. If a low formation of particles nevertheless occurs, the particles which are formed can stick to the protective film and thus do not pass into the critical opening region of the workpiece to be beaded. This is of great importance, in particular, in the case of cartridges for preserving and storing active pharmaceutical ingredients.

[0006] Overall, a high useful life of the individual components of the device for beading a tube end can be achieved by way of the low-wear beading, with the result that large maintenance intervals can be realized. This relates, in particular, to the beading die.

[0007] This leads to it being possible for beading dies to then also be used in a reliable and reproducible manner instead of the frequently used beading rollers. Roller bending tools of this type such as, in particular, beading dies are as a rule less expensive and simpler to handle than a beading roller. Moreover, the requirements of the edge quality of the workpiece to be beaded, in particular with regard to the cut quality, are as a rule lower than in the case of the devices which are already known in the prior art. In particular, the workpiece to be beaded can be machined in a dry state and without re-greasing.

[0008] The protective film can preferably be mounted on an endless reel. In this case, the protective film can be pulled gradually off the endless reel and wound onto an empty reel. In this way, a new piece of the protective film can be used for each beading operation. As a result, particles which adhere to the protective film can be transported away gradually and do not pass into the critical opening region of a following workpiece. In order for it to be possible for the protective film to be wound up and unwound as effectively as possible, at least one pair of transport rollers can be

present between the endless reel and the empty reel. In this case, the transport rollers can preferably be positioned directly upstream of or directly downstream of the receptacle unit. This makes optimum tensioning of the protective film in the region of the receptacle unit and therefore also in the region of the workpiece to be beaded possible.

[0009] The protective film can fundamentally contain, for example, polyethylene (PE), felt, cotton or paper. Preferably, the protective film can consist of a polyvinyl fluoride or can be coated with a polyvinyl fluoride. Protective films of this type have particularly satisfactory sliding and adhesive properties.

[0010] The receptacle unit can preferably be mounted in a longitudinally displaceable manner. As a result, the beading die and the receptacle unit might be moved mutually towards one another and away from one another. The two components are therefore at a greater spacing from the protective film in their starting position, with the result that there is sufficient space for fitting the receptacle units with the workpiece to be beaded. At the same time, the protective film has to be deflected merely slightly out of its path during the beading operation.

[0011] The beading die can preferably have a conical inner contour, with the result that optimum beading results can be realized.

[0012] In the case of the method for beading the tube end, the workpiece to be beaded is first of all positioned in a receptacle unit. A protective film is positioned between this receptacle unit and a beading die which is mounted in a longitudinally displaceable manner. Subsequently, the free edge of the workpiece to be beaded is beaded inwards by means of the beading die. Here, the protective film is situated between that edge of the workpiece which is to be bent over and the beading die. In this way, considerably reduced abrasion occurs, with the result that more exact beading results can be achieved with a longer useful life of the tools, in particular of the beading die. In particular, the beading results remain identical over the entire useful life of the beading die, with the result that optimum beading results are still possible even towards the end of the planned useful life.

[0013] A primary packaging can preferably be positioned in the workpiece to be beaded. In this case, the free edge of the workpiece to be beaded is beaded around the edge of the primary packaging. In this case, in particular, the primary packaging can consist of a plastic material. In this case, for example, the workpiece to be beaded can be configured as an aluminium can or as a steel can.

[0014] The protective film can preferably be unwound from an endless reel. In this way, a new region of the protective film which is still unused can be used for each beading operation. The particles which adhere to the protective film can therefore be transported away from the critical region, with the result that subsequent contamination of the contents present in the workpiece to be beaded cannot occur.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] The invention will be described and explained in greater detail in the following text on the basis of the exemplary embodiment, which is shown in the drawing, in which:

[0016] FIG. 1 shows a diagrammatic side view of the device for beading a tube end before the start of the beading operation,

[0017] FIG. 2 shows a detailed view of the receptacle unit with the workpiece to be beaded and the beading die before the start of the beading operation, and

[0018] FIG. 3 shows a detailed view according to FIG. 2 after the beading of the free edge of the workpiece.

DETAILED DESCRIPTION

[0019] A diagrammatic side view of the device **10** for beading the tube end **12** of a workpiece **14** is

shown in FIG. 1.

[0020] The device **10** has a receptacle unit **20**, in which the workpiece **14** to be beaded is positioned. In the present example, the workpiece **14** to be beaded is a cartridge made from aluminum. In the present example, a primary packaging **22** made from a plastic material is inserted into the workpiece **14**. Here, the primary packaging **22** can be filled with the desired content, as a rule with an active pharmaceutical ingredient. At its end which protrudes out of the receptacle unit **20**, the primary packaging **22** has an open edge **24**, with the result that the primary packaging **22** is configured substantially as an open container. At its end which protrudes out of the receptacle unit **20**, the workpiece **14** has a tube end **12** in the form of a free edge. This tube end **12** of the workpiece **14** projects some way beyond the open edge **24** of the primary packaging **22** (see also FIG. 2). In contrast to the exemplary embodiment which is shown here, the primary packaging **22** might also be dispensed with. In this case, the content to be packaged would be present directly in the workpiece **14**.

[0021] In the present example, the receptacle unit **20** is mounted such that it can be displaced in the longitudinal direction **26** (arrow **28**).

[0022] A beading die **30** which is aligned with the receptacle unit **20** is arranged above the receptacle unit **20**. The beading die **30** is mounted in a displaceable manner in the longitudinal direction **26** (arrow **32**), with the result that the beading die **30** and the receptacle unit **20** can be moved towards one another during the beading operation. On its side which points towards the receptacle unit **20**, the beading die **30** has a conical inner contour **34**, with the result that the free edge **12** of the workpiece **14** can be bent over inwards by means of the beading die **30** (see FIG. 3). In the present example, the free edge **12** of the workpiece **14** is bent over the open edge **24** of the primary packaging **22** here.

[0023] A protective film **40** is positioned between the beading die **30** and the receptacle unit **20**. The protective film **40** is mounted on an endless reel **42** and can be unwound gradually from this endless reel **42**. To this end, in the present example, there are two pairs of transport rollers **44**, **46**. Here, the two transport roller pairs **44**, **46** are positioned close beside the receptacle unit **20**. The protective film **42** can be wound gradually onto an empty reel **48**.

[0024] The protective film **42** can be unwound from the endless reel **42** and wound onto the empty reel **48** cyclically. As a result, a new region of the protective film can be positioned between the beading die **30** and the tube end **12** of the workpiece **14** during each beading operation. In this way, the abrasion on the workpiece **14** and on the conical inner contour **34** of the beading die **30** can be reduced considerably. This leads to a longer useful life of the beading die **30** and therefore also of the device **10**. Moreover, more exact results can be achieved when beading the tube ends **12** of the workpieces **14**.

[0025] Particles which possibly accrue stick to the protective film **40** and are removed from the region of the receptacle unit **20** by way of the further transport of the protective film **40**. As a result, the particles cannot accidentally pass into the interior space of the workpiece **14** or the interior space of the primary packaging **22** and lead to contamination of the content there.

Claims

1. A device (**10**) for beading a tube end (**12**), comprising: a receptacle unit (**20**) for a workpiece (**14**) to be beaded; a beading die (**30**), the beading die (**30**) being mounted in a longitudinally displaceable manner and aligned with the receptacle unit (**20**); and a protective film (**40**) positioned between the beading die (**30**) and the receptacle unit (**20**) for the workpiece (**14**) to be beaded.
2. The device according to claim 1, wherein the protective film (**40**) is mounted on an endless reel (**42**), wherein the protective film (**40**) is pulled off the endless reel (**42**) and wound onto an empty reel (**48**), and wherein the receptacle unit (**20**) and the beading die (**30**) are arranged between the endless reel (**42**) and the empty reel (**48**).

3. The device according to claim 2, wherein a pair of transport rollers (44, 46) is arranged between the endless reel (42) and the empty reel (48).
 4. The device according to claim 3, wherein the pair of transport rollers (44, 46) is positioned directly upstream of or directly downstream of the receptacle unit (20).
 5. The device according to claim 1, wherein the protective film (40) consists of a polyvinyl fluoride or is coated with a polyvinyl fluoride.
 6. The device according to claim 1, wherein the receptacle unit (20) is mounted in a longitudinally displaceable manner.
 7. The device according to claim 1, wherein the beading die (30) has a conical inner contour (34).
 8. A method for beading a tube end (12), comprising: positioning a workpiece (14) to be beaded in a receptacle unit (20); positioning a protective film (40) between the receptacle unit (20) and a beading die (30), the beading die (30) being mounted in a longitudinally displaceable manner; and beading a free edge (12) of the workpiece (14) inwards by the beading die (30).
 9. The method according to claim 8, further comprising positioning a primary packaging (22), in particular a primary packaging (22) made from a plastic material, in the workpiece (14) to be beaded, and beading the free edge (12) of the workpiece (14) around the edge (24) of the primary packaging (22).
 10. The method according to claim 9, wherein the primary packaging (22) is made from a plastic material.
 11. The method according to claim 8, further comprising unwinding the protective film (40) from an endless reel (42).
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